CAMELS-PE: Catchment Attributes and Meteorology for Large-sample Studies – Peru

Technical documentation

Version 1.0.1

1. Introduction

CAMELS-PE (Catchment Attributes and Meteorology for Large-sample Studies – Peru) is a standardized large-sample hydrological dataset that provides hydroclimatic time series, static catchment attributes, metadata, and geospatial information for a representative set of river basins across Peru. The dataset follows the structure and principles of the international CAMELS initiative, providing a consistent framework for hydrological analyses, regionalization studies, environmental assessments, and the development of data-driven hydrological models.

The dataset is organized into four complementary components: metadata, catchment attributes, daily hydroclimatic time series, and geospatial data. These components are distributed using open and widely supported formats, including comma-separated values (CSV) tables and GeoPackage (GPKG) files. All files are linked through a unique catchment identifier (gauge_id), enabling straightforward integration of information across the different dataset components.

This document provides a technical description of the CAMELS-PE dataset and serves as a user guide to its organization and contents. It describes the directory structure, individual data files, variables, units, naming conventions, coordinate reference system, and data formats included in the public release. The objective is to provide users with the information required to understand and efficiently use the dataset. In addition, CAMELS-PE is accompanied by the RCamelsPE package, which provides functions for accessing, querying, visualizing, and downloading the dataset within the R environment.

2. Dataset Overview

CAMELS-PE is organized as a standardized collection of hydrological, meteorological, physiographic, and geospatial data describing river catchments across Peru. The dataset has been designed to provide a consistent and interoperable database for large-sample hydrological studies while maintaining compatibility with other datasets developed under the CAMELS framework. The current release (Version 1.0.1) includes daily hydroclimatic time series, metadata, geospatial layers, and 79 static catchment attributes describing 136 catchments distributed across the Pacific, Atlantic, and Titicaca hydrographic regions of Peru. The dataset contains 10 daily hydroclimatic variables organised under a common daily calendar (1981–2025).

The dataset is distributed as comma-separated values (CSV) files for tabular data and GeoPackage (GPKG) files for geospatial data. These open formats ensure compatibility with commonly used scientific software, including R, Python, QGIS, ArcGIS, and other GIS and data analysis platforms. Table 1 summarizes the main characteristics of the CAMELS-PE dataset.

Table 1. General characteristics of the CAMELS-PE dataset.

Property	Description
Dataset	CAMELS-PE
Version	1.0.1
Country	Peru
Temporal coverage	1981–2025
Temporal resolution	Daily
Formats	CSV and GeoPackage
CRS	WGS84 (EPSG:4326)
License	CC BY 4.0
Catchments	136

3. Dataset Organization

The CAMELS-PE dataset is organized into four main directories that separate metadata, catchment attributes, hydroclimatic time series, and geospatial information. This modular structure simplifies data access while maintaining consistency across all dataset components. Every file is linked through the unique catchment identifier (gauge_id), allowing users to combine information from different directories using standard relational operations. The directory structure of the dataset is shown below.

```
CAMELS-PE/
│
├──── 01_metadata/
│    ├──── stations.csv
│    └──── data_dictionary.csv
│
├──── 02_attributes/
│    ├──── topographic_attributes.csv
│    ├──── climatic_indices.csv
│    ├──── hydrological_signatures.csv
│    ├──── landcover_attributes.csv
│    ├──── geologic_attributes.csv
│    ├──── soil_attributes.csv
│    └──── human_intervention_attributes.csv
│
├──── 03_timeseries/
│    ├──── timeseries.csv
│    └──── by_catchment/
│      ├──── PE_XXXX.csv
│      └──── ...
│
└──── 04_geospatial/
   ├──── camels_pe_catchments.gpkg
   ├──── camels_pe_gauges.gpkg
   └──── by_catchment/
     ├──── PE_XXXX/
     │    ├──── PE_XXXX_catchment.gpkg
     │    └──── PE_XXXX_outlet.gpkg
     └──── ...
```

4. Directory Description

The dataset is divided into four thematic components, each containing a specific type of information required for hydrological analyses.

01_metadata

This directory contains descriptive information for all gauging stations included in CAMELS-PE together with the complete data dictionary. These files provide the information required to identify catchments and interpret every variable distributed within the dataset.

02_attributes

This directory contains static catchment descriptors that do not vary over time. The attributes are grouped into thematic tables describing topography, climate, hydrological behaviour, land cover, geology, soils, and indicators of human intervention. Each table contains one record per catchment identified by gauge_id.

03_timeseries

This directory contains daily hydroclimatic time series for all catchments. The complete database is provided as a single CSV file (timeseries.csv) in long format, while the by_catchment subdirectory contains one file for each individual catchment to facilitate analyses focused on a single basin.

04_geospatial

This directory contains vector representations of catchment boundaries and gauging station locations. General layers including all catchments are provided together with individual GeoPackage files for each catchment, allowing direct integration with geographic information systems and spatial analysis software.

5. Metadata Files

The 01_metadata directory contains the information required to identify and interpret the catchments included in CAMELS-PE. It includes descriptive information for each gauging station and a comprehensive data dictionary describing every variable distributed within the dataset. These files provide the reference information needed to relate catchments, interpret variables, and combine the different dataset components.

The metadata are distributed in two comma-separated values (CSV) files:

- stations.csv, which contains descriptive information for all gauging stations and their corresponding catchments.
- data_dictionary.csv, which provides the technical description of every variable included in the dataset.

Together, these files serve as the primary reference for understanding the organization and contents of CAMELS-PE.

stations.csv

The stations.csv file contains one record for each catchment included in CAMELS-PE and provides the metadata required to uniquely identify gauging stations and their associated drainage basins. The file also documents the temporal completeness of observed streamflow records and the topology of nested catchments. Besides the station identifier and geographic location, the file includes information describing the monitoring period, the percentage of available observations, and the hierarchical relationships among nested catchments. The variable gauge_id is the primary key of the CAMELS-PE dataset and is consistently used throughout all metadata, attribute tables, hydroclimatic time series, and geospatial layers. This common identifier enables users to efficiently integrate information across different dataset components. Table 2 describes the variables contained in stations.csv.

Table 2. Variables included in stations.csv.

Variable	Description
gauge_id	Unique catchment identifier.
gauge_name	Official gauging station name.
gauge_region	Hydrographic region where the station is located.
gauge_lat	Latitude of the gauging station (WGS84).
gauge_lon	Longitude of the gauging station (WGS84).
gauge_elev	Elevation of the gauging station (m a.s.l.).
gauge_record_start	Beginning of the available streamflow record.
gauge_record_end	End of the available streamflow record.
gauge_perc_obs	Percentage of valid daily observed streamflow values during 1981–2025 after quality control.
name_cat	Official catchment name.
is_nested	Indicates whether the catchment belongs to a nested river network.
nested_group_id	Identifier of the downstream catchment within the nested group.
nested_group_size	Number of catchments belonging to the same nested system.
downstream_gauge_id	Identifier of the immediately downstream catchment.
upstream_gauge_id	Identifier of the immediately upstream catchment.

The metadata contained in this file allow users to identify the location and characteristics of each gauging station, evaluate the temporal availability of observations, and identify upstream–downstream relationships among nested catchments.

data_dictionary.csv

The data_dictionary.csv file provides the technical description of every variable included in CAMELS-PE. It serves as the central reference for interpreting the dataset by documenting the content, meaning, and origin of each variable distributed across all files.

Each record in the data dictionary corresponds to a single variable and specifies the dataset component in which it is stored, the corresponding file, its thematic category, the variable name, a concise description, the measurement units (when applicable), and the original data source.

The data dictionary is intended to facilitate the interpretation of the dataset and to ensure consistent use of variable names across different analyses. Table 3 summarizes the information contained in data_dictionary.csv.

Table 3. Variables included in data_dictionary.csv.

Field	Description
folder	Directory containing the file.
file	Name of the file containing the variable.
category	Dataset component or thematic category.
variable	Variable name used in the dataset.
description	Technical description of the variable.
unit	Measurement unit (when applicable).
source	Original data source used to derive the variable.

The data dictionary includes variables from all dataset components, including metadata, catchment attributes, hydroclimatic time series, and geospatial data, providing a single reference for understanding the complete CAMELS-PE dataset.

6. Catchment Attribute Files

The 02_attributes directory contains static descriptors that characterize the physiographic, climatic, hydrological, geological, ecological, and anthropogenic properties of each catchment included in CAMELS-PE. Unlike the hydroclimatic time series, these variables do not vary over time and represent long-term or intrinsic characteristics of the drainage basins.

The attributes are organized into seven thematic tables according to the type of information they describe. Together, these tables contain 79 static catchment attributes computed using a consistent processing workflow. Each file contains one record per catchment identified by the unique identifier gauge_id, allowing straightforward integration with the metadata, time-series, and geospatial components of the dataset. These attributes are intended to support a wide range of applications, including catchment classification, hydrological regionalization, comparative hydrology, rainfall–runoff modelling, machine learning, and environmental analyses. The following sections describe each attribute table included in CAMELS-PE.

topographic_attributes.csv

Topographic characteristics exert a fundamental control on hydrological processes by influencing precipitation distribution, runoff generation, drainage efficiency, and flow routing. The topographic_attributes.csv file summarizes the principal geometric and elevation-related properties of each catchment derived from the FABDEM v1.2 digital elevation model.

Each record corresponds to one catchment and includes basic morphometric descriptors commonly used in hydrological modelling and regionalization studies. Table 4 summarizes the information contained in topographic_attributes.csv.

Table 4. Variables included in topographic_attributes.csv.

Variable	Description
area	Catchment area.
perimeter	Catchment perimeter.
elev_min	Minimum catchment elevation.
elev_max	Maximum catchment elevation.
elev_mean	Mean catchment elevation.
elev_median	Median catchment elevation.
slope_mean	Mean catchment slope.

These variables provide a concise representation of catchment morphology and are useful for characterizing basin size, relief, and terrain gradients.

climatic_indices.csv

The climatic_indices.csv file contains long-term climatic descriptors computed from daily hydroclimatic observations. These indices summarize the average climatic conditions experienced by each catchment and describe precipitation seasonality, aridity, and the occurrence of wet and dry conditions.

The climatic indices provide an overview of the hydroclimatic setting of each basin and facilitate comparative analyses among catchments with contrasting climatic regimes. Table 5 summarizes the information contained in climatic_indices.csv.

Table 5. Variables included in climatic_indices.csv.

Variable	Description
gauge_id	Unique catchment identifier.
p_mean	Mean daily precipitation.
pet_mean	Mean daily potential evapotranspiration.
aridity	Ratio between mean PET and mean precipitation.
p_seasonality	Precipitation seasonality index.
high_prec_freq	Frequency of high-precipitation days.
high_prec_dur	Mean duration of high-precipitation events.
high_prec_timing	Season during which high-precipitation events occur most frequently.
low_prec_freq	Frequency of dry days.
low_prec_dur	Mean duration of dry periods.
low_prec_timing	Season during which dry periods occur most frequently.

hydrological_signatures.csv

Hydrological signatures summarize the long-term response of river catchments and provide quantitative measures of streamflow behaviour. All hydrological signatures were derived from simulated streamflow to ensure temporal consistency across all catchments. The hydrological_signatures.csv file contains a collection of widely used metrics describing runoff generation, flow variability, flow duration characteristics, and low- and high-flow behaviour. These variables are particularly useful for hydrological regionalization, catchment classification, model evaluation, and comparative hydrology. Table 6 summarizes the information contained in hydrological_signatures.csv.

Table 6. Variables included in hydrological_signatures.csv.

Variable	Description
q_mean	Mean daily streamflow.
runoff_ratio	Ratio between streamflow and precipitation.
stream_elas	Streamflow elasticity with respect to precipitation.
baseflow_index	Baseflow index.
slope_fdc	Slope of the flow duration curve.
hfd_mean	Mean half-flow date.
Q5	Fifth percentile of daily streamflow.
Q95	Ninety-fifth percentile of daily streamflow.
high_q_freq	Frequency of high-flow events.
high_q_dur	Mean duration of high-flow events.
low_q_freq	Frequency of low-flow events.
low_q_dur	Mean duration of low-flow events.
zero_q_freq	Percentage of zero-flow days.

landcover_attributes.csv

Land cover influences the partitioning of precipitation into evapotranspiration, infiltration, and runoff, and therefore plays an important role in determining catchment hydrological response. The landcover_attributes.csv file describes the spatial distribution of the dominant land-cover classes within each catchment based on the most recent land-cover information available from the MapBiomas Peru and Ecuador project.

The variables represent the percentage of the catchment occupied by each major land-cover class together with the dominant land-cover category. These descriptors are particularly useful for studies evaluating the influence of vegetation and land use on hydrological processes. Table 7 summarizes the information contained in landcover_attributes.csv.

Table 7. Variables included in landcover_attributes.csv.

Variable	Description
agricul_perc	Percentage of catchment covered by agricultural areas.
forest_perc	Percentage of catchment covered by forest.
non_veget_perc	Percentage of catchment covered by non-vegetated areas.
non_woody_perc	Percentage of catchment covered by non-woody natural vegetation.
water_perc	Percentage of catchment covered by water bodies.
non_forest_perc	Percentage of catchment covered by non-forest natural formations.
non_identi_perc	Percentage of catchment classified as unidentified land cover.
land_dominant_class	Dominant land-cover class within the catchment.

geologic_attributes.csv

Geological characteristics strongly influence groundwater storage, infiltration capacity, and subsurface flow pathways. The geologic_attributes.csv file summarizes the dominant lithological and hydrogeological characteristics of each catchment using information derived from the Global Lithological Map (GLiM) and the GLobal HYdrogeology MaPS (GLHYMPS). The dataset includes the dominant lithological classes together with representative estimates of subsurface porosity and permeability, providing information relevant to groundwater recharge and baseflow generation. Table 8 summarizes the information contained in geologic_attributes.csv.

Table 8. Variables included in geologic_attributes.csv.

Variable	Description
geol_class_1st	Dominant geological class within the catchment.
geol_class_1st_perc	Percentage occupied by the dominant geological class.
geol_class_2nd	Second most abundant geological class.
geol_class_2nd_perc	Percentage occupied by the second dominant geological class.
inter_volca_rocks_perc	Percentage of intermediate volcanic rocks.
geol_porosity	Area-weighted average subsurface porosity.
geol_permeability	Area-weighted average subsurface permeability.

soil_attributes.csv

Soil properties regulate infiltration, water storage, evapotranspiration, and runoff generation. The soil_attributes.csv file describes the dominant soil classes present within each catchment based on the Digital Soil Map (DSOLMap). Rather than providing detailed soil profiles, the dataset summarizes the proportion of the catchment occupied by the principal soil orders together with the dominant soil class. These variables are intended to support regional hydrological analyses and catchment characterization. Table 9 summarizes the information contained in soil_attributes.csv.

Table 9. Variables included in soil_attributes.csv.

Variable	Description
inceptisols_perc	Percentage of Inceptisols.
entisols_perc	Percentage of Entisols.
alfisols_perc	Percentage of Alfisols.
ultisols_perc	Percentage of Ultisols.
aridisols_perc	Percentage of Aridisols.
gelisols_perc	Percentage of Gelisols.
oxisols_perc	Percentage of Oxisols.
mollisols_perc	Percentage of Mollisols.
vertisols_perc	Percentage of Vertisols.
soil_dominant_class	Dominant soil class within the catchment.

human_intervention_attributes.csv

Natural hydrological processes may be substantially modified by human activities such as water abstraction and reservoir operation. The human_intervention_attributes.csv file summarizes indicators describing anthropogenic influences within each catchment using information provided by the National Water Authority of Peru (ANA). The variables quantify the presence of surface-water and groundwater availability accreditations together with the number, storage capacity, and dominant use of reservoirs located within each catchment. These descriptors allow users to distinguish between relatively natural and more strongly regulated basins. Table 10 summarizes the information contained in human_intervention_attributes.csv.

Table 10. Variables included in human_intervention_attributes.csv.

Variable	Description
surface_adh_n	Number of surface-water availability accreditations.
surface_adh_vol	Total accredited surface-water volume.
surface_adh_use	Dominant use of accredited surface water.
groundwater_adh_n	Number of groundwater availability accreditations.
groundwater_adh_vol	Total accredited groundwater volume.
groundwater_adh_use	Dominant use of accredited groundwater.
reservoir_n	Number of reservoirs within the catchment.
reservoir_vol	Total reservoir storage capacity.
reservoir_use	Dominant reservoir use category.

These indicators provide a first-order description of human influence on water resources and may be used to identify catchments affected by water management infrastructure or intensive water use.

7. Time-series Files

The 03_timeseries directory contains the daily hydroclimatic variables included in CAMELS-PE. These data provide the temporal component of the dataset and include both meteorological forcing variables and streamflow records for each catchment. All time series are referenced using the common identifier gauge_id, ensuring compatibility with the metadata, attribute tables, and geospatial data.

To facilitate different analysis workflows, the hydroclimatic data are distributed in two complementary formats. A single master file (timeseries.csv) contains the complete database in long-table format, while the by_catchment directory provides one file for each catchment, allowing users to access individual basins without filtering the master dataset.

The hydroclimatic variables are provided at daily temporal resolution for the period 1981–2025 and include precipitation, precipitation variance, observed and simulated streamflow, potential evapotranspiration, air temperature, solar radiation, and vapor pressure.

The timeseries.csv file contains the complete hydroclimatic database in a long-table format, where each row represents the observations corresponding to one catchment on one calendar day. Although all files follow a common 1981–2025 daily calendar, variables retain the temporal coverage of their original source products, with unavailable dates represented as NA.

Table 11. Variables included in time-series files.

Variable	Description	Unit
date	Calendar date (YYYY-MM-DD)	—
gauge_id	Unique catchment identifier	—
prec	Daily precipitation	mm day^{-1}
prec_var	Daily precipitation variance	mm^2 day^{-2}
flow_obs	Observed streamflow	mm day^{-1}
flow_sim	Simulated streamflow	mm day^{-1}
pet	Potential evapotranspiration	mm day^{-1}
tmin	Minimum air temperature	°C
tmean	Mean air temperature	°C
tmax	Maximum air temperature	°C
srad	Surface solar radiation	MJ m^{-2} day^{-1}
vprp	Vapor pressure	hPa

For users interested in analysing a single basin, CAMELS-PE also provides individual hydroclimatic files within the by_catchment directory.

Each file contains the complete daily record for one catchment and follows the naming convention: PE_XXXX.csv; where XXXX represents the unique catchment identifier.

The structure and variables contained in each individual file are identical to those included in timeseries.csv, differing only in that each file contains data for a single catchment. This organization simplifies analyses focused on individual basins and reduces the need to filter the master database.

All hydroclimatic variables are provided at daily resolution and cover the period from 1 January 1981 to 31 December 2025. The temporal coverage is consistent across the meteorological variables, while the availability of observed streamflow depends on the operating period of each gauging station.

Users should note that the length of observed streamflow records may differ among catchments due to differences in monitoring history and data availability. The beginning and end of the observation period for each station are documented in the stations.csv metadata file.

8. Geospatial Data

The 04_geospatial directory contains the vector data associated with the CAMELS-PE dataset. These files provide the spatial representation of gauging stations and catchment boundaries, allowing users to integrate hydroclimatic information with geographic analyses and geographic information systems (GIS).

All spatial data are distributed in GeoPackage (GPKG) format, an open standard developed by the Open Geospatial Consortium (OGC). Compared with the traditional ESRI Shapefile format, GeoPackage supports larger datasets, longer attribute names, Unicode characters, and multiple vector layers within a single file while avoiding many of the limitations associated with shapefiles. All geospatial layers use the World Geodetic System 1984 (WGS84) geographic coordinate reference system (EPSG: 4326). The file camels_pe_catchments.gpkg contains the polygon boundaries of all catchments included in CAMELS-PE.

Each polygon represents the drainage area associated with a streamflow gauging station and is uniquely identified by gauge_id, allowing direct linkage with the metadata, attribute tables, and hydroclimatic time series.

The GeoPackage contains one polygon feature for each catchment included in the dataset. The file camels_pe_gauges.gpkg contains the outlet locations of all streamflow gauging stations included in CAMELS-PE. Each point corresponds to the outlet of a catchment and shares the same gauge_id used throughout the dataset. The geographic coordinates are expressed in latitude and longitude using WGS84.

In addition to the complete geospatial layers, CAMELS-PE provides individual GeoPackage files for each catchment within the by_catchment directory. Each catchment folder contains two files:

- PE_XXXX_catchment.gpkg, containing the catchment polygon.
- PE_XXXX_outlet.gpkg, containing the outlet point.

where XXXX corresponds to the unique catchment identifier.

Table 12 summarizes the geospatial products distributed within CAMELS-PE.

Table 12. Geospatial files included in CAMELS-PE.

File	Geometry	Description
camels_pe_catchments.gpkg	Polygon	Catchment boundaries for all basins.
camels_pe_gauges.gpkg	Point	Outlet locations of all gauging stations.
PE_XXXX_catchment.gpkg	Polygon	Individual catchment boundary.
PE_XXXX_outlet.gpkg	Point	Individual catchment outlet location.

9. Data Formats and Conventions

To ensure consistency and facilitate data integration across all dataset components, CAMELS-PE follows a set of standardized conventions regarding file formats, variable names, identifiers, units, and missing values. These conventions have been adopted consistently throughout the dataset to improve interoperability with commonly used hydrological and geospatial analysis software. Variable names are expressed using concise English abbreviations following conventions commonly adopted within the CAMELS family of datasets. Table 13 summarizes the abbreviations used for the principal hydroclimatic variables in CAMELS-PE.

Table 13. Description of hydroclimatic variables abbreviations.

Variable	Description
prec	Daily precipitation
pet	Potential evapotranspiration
flow_obs	Observed streamflow
flow_sim	Simulated streamflow
tmin	Minimum air temperature
tmean	Mean air temperature
tmax	Maximum air temperature
srad	Surface solar radiation
vprp	Vapor pressure

Missing observations are represented using the standard NA notation in all tabular files. No additional placeholder values (e.g., −9999 or −999) are used within the dataset.

Physical variables are reported using standard hydrological and meteorological units to ensure consistency across the dataset. Table 14 summarizes the units of the principal hydroclimatic variables in CAMELS-PE.

Table 14. Units of the principal hydroclimatic variables.

Variable	Unit
Precipitation	mm day^{-1}
Potential evapotranspiration	mm day^{-1}
Streamflow	mm day^{-1}
Air temperature	°C
Solar radiation	MJ m^{-2} day^{-1}
Vapor pressure	hPa
Catchment area	km^2
Elevation	m a.s.l.

10. Data Sources

The CAMELS-PE dataset integrates information derived from multiple national and international datasets describing meteorological conditions, streamflow, topography, land cover, geology, soils, and human intervention. These datasets were selected according to their spatial coverage, temporal consistency, and suitability for hydrological applications in Peru. Table 15 summarizes the primary datasets from which CAMELS-PE variables were derived.

Table 15. Primary data sources used in CAMELS-PE.

Dataset	Variables	Provider
PISCOp v2.1	Daily precipitation	SENAMHI
PISCOt v1.2	Daily minimum, mean and maximum air temperature	SENAMHI
PISCOeo_pm v1.0	Potential evapotranspiration	SENAMHI
ERA5-Land	Solar radiation, vapor pressure	ECMWF
PISCO_ARNOVIC v1.1	Simulated streamflow	SENAMHI
SENAMHI	Observed streamflow	SENAMHI
FABDEM v1.2	Topographic variables	University of Bristol
MapBiomas Peru and Ecuador	Land cover	MapBiomas Network
GLiM	Lithology	GLiM Consortium
GLHYMPS v2.0	Hydrogeological properties	GLHYMPS Project
DSOLMap	Soil classes	DSOLMap Project
ANA	Reservoirs and water use	National Water Authority (Peru)

11. License

CAMELS-PE is distributed under the terms of the Creative Commons Attribution 4.0 International (CC BY 4.0) license. Users are free to copy, redistribute, adapt, and build upon the dataset provided that appropriate credit is given to the original authors.

12. Contact

For questions, bug reports, or suggestions regarding CAMELS-PE, please contact:

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- R package: https://github.com/hllauca/RCamelsPE